Mahir Al Kamal

Software Engineer — React, Next.js & Node.js

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- Almost four years building React and Node.js apps for government and enterprise clients
- Worked on national platforms for the Bangladesh Election Commission and the Anti-Corruption Commission
- Built an AI grading tool on my own, with a model I trained, now piloted in a university class
- Mentored two interns who were later hired full-time

SKILLS

Languages	JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3
Backend	Node.js, Express, FastAPI, REST APIs, GraphQL, WebSockets, Socket.IO, zod validation
Frontend	React, Next.js (App Router), Redux Toolkit, RTK Query, Tailwind CSS, shadcn/ui, Vite, PWA
Databases	PostgreSQL, MongoDB, DynamoDB, schema design, versioned migrations
AI	Gemini API (vision) in production code, PyTorch model training, ONNX, RAG and embeddings (research), Claude Code, MCP
Security	JWT, OAuth 2.0 / OIDC (AWS Cognito), RBAC, OWASP Top 10, rate limiting, input validation, security audits
DevOps & cloud	Docker, GitHub Actions, AWS (Lambda, API Gateway, SQS, EventBridge, CDK), Vercel, Render, Supabase
Testing	Vitest, React Testing Library, Playwright, pytest, node:test

SUMMARY

Most of my work has been React on the frontend and Node.js on the backend, with PostgreSQL underneath. Part of that was Neustring, an analytics product I worked on with a US client team across time zones. I also write Python: my exam-marks scanner runs on a FastAPI backend with a model I trained in PyTorch. On the side, I teach programming, data structures and algorithms to undergraduate students.

EXPERIENCE

Penta Global Limited, Dhaka

Oct 2023 – Present

Software Engineer

- Turned business-analyst requirements into features with the backend team, across three government and enterprise products
- Moved old Jinja and jQuery screens to React one screen at a time, so feature work never had to stop
- Mentored two interns, both later hired full-time, through pairing and code reviews, and ran sessions on JavaScript fundamentals

PROJECTS AT PENTA GLOBAL LIMITED

Neustring · B2B telecom roaming analytics product for a US-based client

- Built the tables, charts and filtered exports analysts use every day on large roaming datasets, and kept them quick with server-side pagination and memoized selectors
- Set up Amazon QuickSight dashboards next to the app for one-off questions, and worked with the client's US team across time zones

Election Management System & Online Nomination System · Bangladesh Election Commission

- Step-by-step nomination forms that stop a filing from moving forward while its data is incomplete or contradictory
- Role-based screens for candidates, returning officers and admins, with the same rules enforced by the API and every action traceable for audit

ACC ERP · Anti-Corruption Commission

- Role-gated ERP modules and long multi-step forms on REST APIs, with server errors shown on the right field and drafts saved

Independent University, Bangladesh Dec 2022 – Present

Adjunct Faculty & Research Assistant Jan 2023 – Jan 2025; Apr 2026 – Present

- Teach the Algorithms and Data Structures labs in Python, and work on research in LLM-based algorithm selection

Part-Time Frontend Developer Dec 2022 – Sept 2023

- Worked on the frontend of IUB's public website alongside coursework

PROJECTS

Exam Script Mark Scanner · AI grading tool, piloted in a live university class

Faculty photograph exam script cover pages; the app reads student IDs and marks and exports the session to Excel.

- React and TypeScript PWA with a FastAPI backend. OpenCV finds and straightens the marks grid, and the teacher confirms every value before it's saved
- Recognition runs on a local CNN or Tesseract with a Gemini Vision fallback behind one interface, with OpenCV handling detection and segmentation
- Trained the CNN in PyTorch and made it the default after measuring it on real class photos. Student ID images never leave the device
- Hosted on AWS with upload checks, rate limiting and an origin guard. 259 backend and 383 frontend tests; two code audits found 51 issues and 46 are fixed

React, TypeScript, Python, FastAPI, OpenCV, PyTorch, Gemini API, Docker, AWS Lambda, pytest
github.com/iammahir2020/marks-upload

Battle of Polashi · live real-time multiplayer game

An unofficial online version of Polashi, a hidden-role social deduction board game by Playground Inc., built so our group could play from their own phones.

- Node.js/Express and Socket.IO server that owns every game rule. Each event is validated with zod, and each player only receives their own view of the game
- Added per-seat reconnect tokens, rate limits per socket and per IP, origin checks and message size caps after auditing the app as an attacker
- Moved game logs from Firestore to PostgreSQL with check constraints and versioned migrations, and saved live rooms so a deploy doesn't end a game. About 70 backend and 240 frontend tests plus Playwright end-to-end runs

Node.js, Express 5, Socket.IO, PostgreSQL (Supabase), zod, React 19, TypeScript, Vitest, Playwright, GitHub Actions
github.com/iammahir2020/polashi_game_backend · github.com/iammahir2020/polashi_game_frontend

DSE Market Monitor · stock alerts for the Dhaka Stock Exchange

- Investors set price, percentage or volume alerts. A Node.js/Express and MongoDB service checks them every two minutes and notifies users over WebSockets and Telegram, with phone OTP login, JWT sessions and rate limits

Node.js, Express, MongoDB, WebSockets, Telegram Bot API, React, TypeScript
github.com/iammahir2020/dse-monitor-backend · github.com/iammahir2020/dse-monitor-frontend

Event-Driven Backend on AWS · learning project

- Built in AWS CDK with Cognito-protected REST APIs, DynamoDB Streams pushing every write to open browsers through AppSync, and an EventBridge-to-SQS path with a dead-letter queue

AWS CDK, Lambda, API Gateway, DynamoDB, AppSync, EventBridge, SQS, Cognito, TypeScript
github.com/iammahir2020/bff-practice

EDUCATION

MSc in Computer Science · Independent University, Bangladesh Expected 2027

BSc in Computer Science and Engineering · Independent University, Bangladesh · CGPA 3.67 / 4.00 2021

RESEARCH

LLM-Based Algorithm Selection for Tabular Medical Datasets · IUB · manuscript in preparation

- Testing whether LLMs pick ML algorithms for medical datasets by reasoning or by memory. I built the metadata extractor and algorithm cards, and designed the contamination tests

LANGUAGES & REFERENCES

English: Proficient · Bangla: native

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